Part 2 : Gradient Descent Manual calculation

Step 1 :

Given : $\hat{y} = m_1 x_1 + m_2 x_2 + b$

where : Initial $m = [-1, 2]$

Initial $b = [1, 1]$

Data points.

$X = ((2,3),(4,10))$ $y = (5,6)$

$X = \begin{bmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{bmatrix}$, $m = \begin{bmatrix} m_1 \\ m_2 \end{bmatrix}$, $b = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$ $y = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$

Form

$X = \begin{bmatrix} 2 & 3 \\ 4 & 10 \end{bmatrix}_{2\times2}$ $m = \begin{bmatrix} -1 \\ 2 \end{bmatrix}_{2\times1}$, $b = \begin{bmatrix} 1 \\ 1 \end{bmatrix}_{2\times1}$, $y = \begin{bmatrix} 5 \\ 6 \end{bmatrix}_{2\times1}$

Number of observations :

$\hat{y} = xm + b_1 = \begin{bmatrix} 2 & 3 \\ 4 & 10 \end{bmatrix}\begin{bmatrix} m_1 \\ m_2 \end{bmatrix} + \begin{bmatrix} b \\ b \end{bmatrix} = \begin{bmatrix} m_1(2) + m_2(3) + b \\ m_1(4)\, m_2(10) + b \end{bmatrix}$

step 2 : (MSE cost function)

$$f(m,b) = \frac{1}{N} \sum_{i=1}^{N} (\hat{y}(i) - y(i))^2$$

Error column vector (e) : $e = \hat{y} - y = (xm + b) - y$

$$f(m,b) = \frac{1}{N} e^T \cdot e$$

$e^T \rightarrow$ e transpose

$e^T \cdot e$ is scalar product

let $J = \frac{1}{N} \sum_{i=1}^{N} e_i^2$, $e_i = (m_1 x_{i1} + m_2 x_{i2} + b - y^{(i)})$

Taking partial derivative of f with respect to $m_j$ where $j \in \{1, 2\}$,

$$\frac{dJ}{dm_j} = \frac{1}{N} \sum_{i=1}^{N} \frac{d(e_i)^2}{dm_j}$$, let $e_i = u$

by Power Rule and chain Rule:

$$= \frac{d(u^2)}{dm_j} = 2u \cdot \frac{du}{dm_j}$$

$$= \frac{d(e_i)^2}{dm_j} = 2e_i \cdot \frac{de_i}{dm_j}$$

solve for $\frac{de_i}{dm_j}$

$$\frac{de_i}{dm_j} = \frac{d}{dm_j}(m_1 x_{i1} + m_2 x_{i2} + b - y^{(i)}) = x_{ij}$$

so: $$\frac{dJ}{dm_j} = \frac{2}{N} \sum_{i=1}^{N} e_i \cdot x_{ij} = \frac{2}{N} \sum_{i=1}^{N} x_{ij} (\hat{y}^{(i)} - y^{(i)})$$

$$\nabla_{m_j} = \begin{bmatrix} \frac{dJ}{dm_1} \\ \frac{dJ}{dm_2} \end{bmatrix}$$

3: Derivation of J with respect to b

$$\frac{dJ}{db} = \frac{1}{N} \sum_{i=1}^{N} 2e_i \cdot \frac{de_i}{db}$$

$$\frac{de_i}{db} = d(m_1 x_{i1} + m_2 x_{i2} + b - y^{(i)}) = 1$$

$$\frac{dJ}{db} = \frac{2}{N} \sum_{i=1}^{N} e_i = \frac{2}{N} \sum_{i=1}^{N} (\hat{y}^{(i)} - y^{(i)})$$

C, parameter Update formula

$$m^{(k+1)} = m^k - \alpha \Delta m_j^{(k)} \quad \text{where (k) is old}$$

$$b^{(k+1)} = b^k - \alpha \frac{dJ}{db}^{(k)}$$

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error dropped from 61.00 to 6.845

Iteration 2:

Step 1 compute gradient.

$$\frac{dJ}{dm_1} = \frac{2}{N}\sum_{i=1}^{N} e_i \cdot x_{ij} = \frac{2}{2}(-2.96)(1) + (-2.22)(4) = -11.84$$

$$\frac{dJ}{dm_2} = \frac{2}{2}((-2.96)(3) + (-2.22)(10)) = -31.08$$

$$\Delta_m J^{(1)} = \begin{bmatrix} -11.84 \\ -31.08 \end{bmatrix}$$ gradient with respect to m.

No gradients with respect to b

$$\frac{dJ}{db} = \frac{2}{N}\sum_{i=1}^{N} e_i = \frac{2}{2}(-2.96 + (-2.22)) = -5.18$$

Step 2 : Update parameter m

$$m^{(2)} = m^{(1)} - \alpha \nabla_m J^{(1)}$$

$$m^{(2)} = \begin{bmatrix} 1.45 \\ 0.87 \end{bmatrix} - 0.01 \begin{bmatrix} -11.84 \\ -31.08 \end{bmatrix} = \begin{bmatrix} -2.3316 \\ 1.1808 \end{bmatrix}$$

Step 3 Update b

$$b^{(2)} = b^{(2)} - \alpha \frac{dJ^{(1)}}{db} = 0.88 - 0.01(-5.18) = 0.88 + 0.0818 = 0.9318$$

checking changes

$$\hat{y}^{1.(2)} = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} -1.3316 \\ 1.1808 \end{bmatrix} + \begin{bmatrix} 0.9318 \\ 0.9318 \end{bmatrix} = \begin{bmatrix} 3.1426 \\ 7.4134 \end{bmatrix}$$

$$e^{(2)} = \begin{bmatrix} 3.1426 \\ 7.4134 \end{bmatrix} - \begin{bmatrix} 5 \\ 6 \end{bmatrix} = \begin{bmatrix} -1.8574 \\ 1.4134 \end{bmatrix}$$

$$J^{(2)} = \frac{1}{2}((-1.8574)^2 + (2.4134)^2 = 2.7237995$$

Error dropped again from 6.845 to 2.7237995

## Step 3: Iterations

1. Initial prediction:

$$\hat{y}^{(0)} = \begin{bmatrix} 2 & 3 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 17 \end{bmatrix}$$

2. Initial error:

$$e^{(0)} = \begin{bmatrix} 6 \\ 17 \end{bmatrix} - \begin{bmatrix} 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 1 \\ 11 \end{bmatrix} = \begin{bmatrix} e_1 \\ e_2 \end{bmatrix}, \quad J^{(0)} = \frac{2^2 + 11^2}{2} = \underline{61.00}$$

$$\frac{dJ}{dm_1} = \frac{2}{N} \sum_{i=1}^{N} e_i \cdot x_{i,j} = \frac{2}{2} ((1)(1) + (11)(4)) = 45$$

$$\frac{dJ}{dm_2} = \frac{2}{2} ((1)(3) + (11)(10)) = 3 + 110 = 113$$

$$\Delta_m J^{(0)} = \begin{bmatrix} 45 \\ 113 \end{bmatrix}, \quad \frac{dJ^{(0)}}{db} = 1 + 11 = 12$$

### Iteration 1:

let $\alpha = 0.01$

$$m^{(1)} = \begin{bmatrix} -1 \\ 2 \end{bmatrix} - 0.01 \begin{bmatrix} 45 \\ 113 \end{bmatrix} = \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix}$$

$$b^{(1)} = 1 - 0.01(12) = 0.88$$

$$m^{(1)} = \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix}, \quad b^{(1)} = \begin{bmatrix} 0.88 \\ 0.88 \end{bmatrix}$$

$$\hat{y}^{(1)} = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix} + \begin{bmatrix} 0.88 \\ 0.88 \end{bmatrix} = \begin{bmatrix} 2.04 \\ 3.78 \end{bmatrix}$$

$$e^{(1)} = \begin{bmatrix} 2.04 \\ 3.78 \end{bmatrix} - \begin{bmatrix} 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 2.04 - 5 \\ 3.78 - 6 \end{bmatrix} = \begin{bmatrix} -2.96 \\ -2.22 \end{bmatrix}$$

$$J^{(1)} = \frac{1}{2} [(-2.96)^2 + (-2.22)^2 = \underline{6.845}$$

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# Iteration 2

**Step 1** Compute gradient

$$\frac{\partial J}{\partial m_1} = \frac{2}{N}\sum_{i=1}^{N} e_i \cdot x_{ij} = \frac{2}{2}\left((-2.96)(1) + (-2.22)(4)\right) = -11.84$$

$$\frac{\partial J}{\partial m_2} = \frac{2}{2}\left((-2.96)(3) + (-2.22)(10)\right) = -31.08$$

$$\nabla_m J^{(1)} = \begin{bmatrix} -11.84 \\ -31.08 \end{bmatrix}, \text{ gradient with respect to } m$$

No gradient with respect to b

$$\frac{\partial J}{\partial b} = \frac{2}{N}\sum_{i=1}^{N} e_i = \frac{2}{2}\left(-2.96 + (-2.22)\right) = -5.18$$

**Step 2**: Update parameter m

$$m^{(2)} = m^{(1)} - \alpha \nabla_m J^{(1)}$$

back

$$m^{(2)} = \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix} - 0.01 \begin{bmatrix} -12.84 \\ -31.08 \end{bmatrix} = \begin{bmatrix} -2.3316 \\ 1.1808 \end{bmatrix}$$

Step 3: Update b

$$b^{(2)} = b^{(1)} - \alpha \frac{\partial J^{(1)}}{db}$$

$$b^{(2)} = 0.88 - 0.01(-5.18) = 0.88 + 0.0518 = \underline{0.9318}$$

Checking Changes

$$y^{(2)} = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \begin{pmatrix} -1.3316 \\ 1.1808 \end{pmatrix} + \begin{pmatrix} 0.9318 \\ 0.9318 \end{pmatrix} = \begin{bmatrix} 3.1426 \\ 7.4134 \end{bmatrix}$$

$$e^{(2)} = \begin{bmatrix} 3.1426 \\ 7.4134 \end{bmatrix} - \begin{bmatrix} 5 \\ 6 \end{bmatrix} = \begin{bmatrix} -1.8574 \\ 1.4134 \end{bmatrix}$$

$$J^{(2)} = \frac{1}{2}\left((-1.8574)^2 + (1.4134)^2 = \underline{2.7237995}\right.$$

Error highly dropped from 6.845 to 2.723799

## Iteration 3

* Use parameters and gradient from iteration 2

**Step 1: Calculate gradient**

$$\nabla_m J = \begin{bmatrix} \frac{\partial J}{\partial m_1} \\ \frac{\partial J}{\partial m_2} \end{bmatrix}, \quad \nabla_b J = \frac{\partial J}{\partial b}$$

$$\frac{\partial J}{\partial m_j} = \frac{2}{N} \sum_{i=1}^{N} e_i \cdot x_{ij}, \quad \frac{\partial J}{\partial b} = \frac{2}{N} \sum_{i=1}^{N} e_i$$

$$\frac{\partial J}{\partial m_1} = \frac{2}{2}\left((-2.8574)(1) + (1.4134)(4)\right) = 3.7962$$

$$\frac{\partial J}{\partial m_2} = \frac{2}{2}\left((-2.8574)(3) + (1.4134)(10)\right) = 8.5618$$

$$\Rightarrow \nabla_m J = \begin{bmatrix} 3.7962 \\ 8.5618 \end{bmatrix}, \quad \nabla_b J = 0.444$$

$$\frac{\partial J}{\partial b} = \frac{2}{N} \sum_{i=1}^{N} e_i = \frac{2}{2}\left((-2.8574) + (1.4134)\right) = 0.444$$

**Step 2: Update parameters:**

$$m^{(2)} = m^{(1)} - \alpha J^{(1)}, \quad b^{(2)} = b^{(1)} - \alpha J^{(1)}$$

$$m^{(2)} = \begin{bmatrix} -2.3316 \\ 1.1802 \end{bmatrix} - 0.01 \begin{bmatrix} 3.7962 \\ 8.5628 \end{bmatrix} = \begin{bmatrix} -1.369562 \\ 1.095182 \end{bmatrix}$$

$$b^{(2)} = 0.9318 - 0.01(0.444) = 0.92736$$

**Step 3: Checking changes**

$$\hat{y}^{(3)} = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} -1.369562 \\ 1.095182 \end{bmatrix} + \begin{bmatrix} 0.92736 \\ 0.92736 \end{bmatrix} = \begin{bmatrix} 2.843344 \\ 5.473572 \end{bmatrix}$$

$$e^{(3)} = \begin{bmatrix} 2.843344 \\ 5.473572 \end{bmatrix} - \begin{bmatrix} 5 \\ 6 \end{bmatrix} = \begin{bmatrix} -2.156656 \\ -0.526428 \end{bmatrix}$$

$$J^{(3)} = \frac{1}{2}\left((-2.156656)^2 + (-0.526428)^2\right)$$

$$J^{(3)} = 2.46415$$

Summary: After 3 iterations, cost dropped from 61.00 to 2.46415

- The updated parameters are mentioned including predictions and errors.
- All show that the algorithm is moving in favor of optimization, approaching 0.